Fiscal Year:	FY 2011	**Task Last Updated:**	FY 08/29/2011
PI Name:	Cooper, Maya M.S.		
Project Title:	Development of Spaceflight Foods with High Microbial Concentrations		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	maya.cooper@nasa.gov	**Fax:**	FY
PI Organization Type:	NASA CENTER	**Phone:**	281.483.1892
Organization Name:	Leidos/NASA Johnson Space Center		
PI Address 1:	1300 Hercules MC:C09		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	09/08/2010	**End Date:**	09/01/2011
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Sullivan, Thomas	**Contact Phone:**	
Contact Email:	thomas.a.sullivan@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/01/2011 per HRP Master Task List dtd 11/11/2011 (Ed., 1/5/2012) NOTE: Start date is actually 9/8/2010, per M. Covington/JSC (Ed., 8/10/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Oubre, Cherie (NASA Johnson Space Center/Wyle) Elliott, Todd (NASA Johnson Space Center/Wyle) Catauro, Patricia (NASA Johnson Space Center/Lockheed Martin)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Strict microbiological standards have been established for the thermostabilized, freeze dried, low moisture, and intermediate moisture foods that make up the current spaceflight food system. However, for long duration missions, it may be prudent to consider space food items with higher microbial counts. The feasibility of including shelf-stable probiotics and establishing appropriate handling procedures for extraterrestrially grown and processed foods needs to be determined in order to re-evaluate microbiological guidelines for the spaceflight food system. A space habitat food system that involves growing and processing fresh produce in situ (bioregenerative foods) and launching payloads of unprocessed grain, legumes, and seeds (bulk foods) is being explored as an avenue to longer shelf life. However, these foods could contain a high concentration of microorganisms with respect to current microbiological standards for microbial load of spaceflight foods. With mounting evidence of a link between the consumption of probiotic microorganisms and health benefits, including increased immune function and cancer prevention, the inclusion of probiotics in the spaceflight food system is also being considered. The potential to impact crew member health through functional foods is yet unexplored but highly relevant to risks associated with space travel. As with extraterrestrial food processing, probiotic foods and supplements fall outside of the current microbiological standards for spaceflight. The purpose of this study is to perform a review of relevant information to assess the feasibility of developing a microbially-safe, spaceflight food system that includes extraterrestrial processing and/or probiotics. If deemed feasible, guidance for standards of microbial acceptability and appropriate testing methods for foods/supplements will be recommended. The study will proceed as follows: 1. Determine the potential of microbial load from bioregenerative produce. 2. Determine the current microbial strains and uses of live cultures in shelf stable foods. 3. Compare levels of microbial load in these foods/supplements to current requirements and assess the level of additional risk. 4. Investigate current literature and commercial operations to provide guidance in establishing microbiological requirements and testing methods for foods with high and diversified microbial concentrations. The final report will recommend future work for the further development of an expanded, safe food system which includes diversified microbial concentrations.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	None at the information review stage.
Task Progress:	Current spaceflight food requirements are inadequate for microbial monitoring associated with crop production, the use of bulk foods, and subsequent processing and handling, as well as foods with inherently high microbial concentrations, such as probiotics, which may be beneficial for crew health. This review provided an initial investigation into (a) the potential of microbial load from bioregenerative produce, (b) the current microbial strains and uses of live cultures in shelf stable foods, (c) a comparison of levels of microbial load in these foods/supplements to current requirements and an assessment of additional risk, and (d) an investigation of current literature and commercial operations to provide guidance in establishing microbiological requirements and testing methods for foods with high and diversified microbial concentrations. The evidence from this review indicates a need for further research to determine the procedures necessary for growing and processing regenerative produce and development of specific cleaning methods to mitigate microbiological risks. In addition, data mining performed during this review identified potential benefits for the use of food products with high microbial concentrations. Specifically, this review has performed an extensive market survey to generate a comprehensive list of potential probiotic products, which should be evaluated in future research to determine preferred probiotic formulations for spaceflight. Future research would also determine the specific microbial testing procedures for those formulations and assess the risk of adding the formulation to the food system.
Bibliography Type:	Description: (Last Updated: 04/23/2019)